

Right-Triangle Trigonometry



Trigonometric ratios

- 1 A right triangle has legs of length 5 and 12 and hypotenuse 13. For the angle θ opposite the side of length 5, find:

a $\sin \theta$ **b** $\cos \theta$ **c** $\tan \theta$
 - 2 Write the exact value of each expression:

a $\sin 30^\circ$ **b** $\cos 45^\circ$ **c** $\tan 60^\circ$ **d** $\sin 60^\circ$
 - 3 A right triangle has legs of length 7 and 24.

a Find the length of the hypotenuse.
b Find the measure of the angle opposite the side of length 7, to the nearest hundredth of a degree.
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Applications

- 4 A ladder leans against a wall, making an angle of 62° with the ground. The foot of the ladder is 4.5 m from the base of the wall. How high up the wall does the ladder reach? Round to two decimal places.
 - 5 From the top of a cliff 50 m above sea level, the angle of depression to a boat is 18° . How far is the boat from the base of the cliff? Round to one decimal place.
 - 6 A kite string is 85 m long and makes an angle of 54° with the horizontal ground. Assuming the string is straight, find the height of the kite above the ground, to the nearest metre.
 - 7 Two buildings are 40 m apart. From the top of the shorter building (height 25 m), the angle of elevation to the top of the taller building is 32° . Find the height of the taller building, to one decimal place.
 - 8 A surveyor stands 120 m from the base of a tower and measures the angle of elevation to the top as 37° . Find the height of the tower, to the nearest tenth of a metre.
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Solving right triangles

- 9 In right triangle ABC with the right angle at C , $A = 35^\circ$ and the hypotenuse $AB = 20$. Find:
- a side BC (opposite A)
 - b side AC (adjacent to A)
 - c angle B
- 10 In right triangle PQR with the right angle at R , the two legs are $PR = 9$ and $QR = 12$. Find angle P (opposite QR) to the nearest tenth of a degree, and the length of the hypotenuse.
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Angles of elevation and depression

- 11 A drone hovers at a height of 60 m. The angle of depression from the drone to a landmark on the ground is 27° . Find the horizontal distance from the point directly below the drone to the landmark, to the nearest metre.
- 12 From a lighthouse 45 m tall, the angle of depression to a ship is 12° , and a second ship is spotted further out with angle of depression 7° . Find the distance between the two ships, to the nearest metre.
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The reciprocal trig ratios

- 13 For a right triangle with opposite = 3, adjacent = 4, hypotenuse = 5 (relative to angle θ), find:
- a $\csc \theta$
 - b $\sec \theta$
 - c $\cot \theta$
- 14 Explain why $\csc \theta = \frac{1}{\sin \theta}$, and use this to find $\csc(30^\circ)$.
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Multi-step surveying problems

- 15 A surveyor at point A measures the angle of elevation to a mountain peak as 22° . Walking 500 m directly toward the mountain to point B , the angle of elevation increases to 35° . Find the height of the mountain, to the nearest metre. (Hint: let h be the height and x the distance from B to the base; set up two equations.)
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The 30-60-90 and 45-45-90 special triangles

- 16 In a 30-60-90 triangle, the shortest side (opposite the 30° angle) has length 5. Find the lengths of the other two sides in exact form.
- 17 In a 45-45-90 triangle, each leg has length 7. Find the hypotenuse in exact form.

Trigonometric ratios beyond the first quadrant (preview)

- 18** A right triangle used to define $\sin \theta$ and $\cos \theta$ only works directly for acute angles θ . Explain why a different (coordinate-based) definition is needed to extend these ratios to obtuse or negative angles.